

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. We prove source-dependent microcanonical preparation for regular interior constraint targets satisfying an explicit affine-span condition. Typical compound-Poisson activity gives the uniform covariance, while linearly many regenerative full-rank blocks yield a globally smooth Fourier component and an exponentially small nonsmoothing remainder. The exact finite-volume saddle then gives the mixed lattice-continuous shell coefficient and prepared rate.

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1. ROUND-TEN RECONSTRUCTION: INTERIOR SOURCE SADDLES AND A REGENERATIVE MIXED-SHELL COEFFICIENT

Let

$$C(z) = \left(1, v, \frac{|v|^2}{2}, \chi_1(z), \dots, \chi_d(z)\right)$$

be the complete initial constraint mark. We impose the model's necessary nondegeneracy in its intrinsic form: the affine span of C on the positive activity support is the full declared constraint affine space. Equivalently, no nonzero linear combination of the added χ_j is almost surely a combination

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of particle number, momentum and energy. The target a lies in a compact subset of the relative interior of the mean image. These are exactly the conditions under which a finite source saddle and a subexponential local coefficient can exist.

1.1. The full finite-volume saddle. For a bounded path/contact source H , define

$$Q_\varepsilon(H, \lambda) = \mu_\varepsilon^{-1} \log \mathbb{E}_\varepsilon^{\text{gc}} e^{\mu_\varepsilon \langle H, X_\varepsilon \rangle + \lambda \cdot C_{\text{tot}}}.$$

Lemma 1.1 (Typical-sector covariance). *On compact source and multiplier charts,*

$$cI \leq D_{\lambda\lambda}^2 Q_\varepsilon(H, \lambda) \leq CI$$

on the declared affine constraint space, uniformly in small ε . The lower bound comes from linearly many particles in disjoint activity boxes, not from an event with a fixed particle number.

Proof. Choose finitely many positive-activity phase boxes whose centered mark vectors span the affine constraint space. Divide the physical torus into $c\mu_\varepsilon$ mutually separated insertion cells. The hard-core cluster expansion gives a conditional occupancy probability bounded away from zero and one for a positive fraction of those cells, uniformly on the source chart. Conditional variance in the spanning mark directions is therefore at least $c\mu_\varepsilon$; division by the pressure speed gives the lower bound. The upper bound follows from the uniform second connected cumulant estimate. Exponentially rare fixed-particle sectors are never used. $\square$

Corollary 1.2 (Exact source-dependent saddle). *For every regular (H, a) there is a unique finite-volume multiplier $\lambda_{\varepsilon, H, a}$ satisfying*

$$D_\lambda Q_\varepsilon(H, \lambda_{\varepsilon, H, a}) = a.$$

It converges locally uniformly, with source derivatives, to the unique limiting multiplier $\lambda_{H, a}$.

Proof. Apply the quantitative inverse-function theorem using Lemma 1.1, normal convergence of the full connected pressure, and compactness of the interior target set. $\square$

1.2. Regenerative smoothing of the complete compound law. A single mark may live on a curved lower-dimensional surface. We therefore use blocks. Fix m_0 disjoint insertion cells and call a block good when exactly one particle lies in each of a selected set of cells and their phase points fall in product patches on which the sum map

$$(z_1, \dots, z_{m_0}) \mapsto \sum_{j=1}^{m_0} C(z_j)$$

has Jacobian rank equal to the constraint dimension. The affine-span condition and the constant-rank theorem provide finitely many such patches.

Lemma 1.3 (Good-block minorization with exponentially small complement). *The configuration contains $c\mu_\varepsilon$ disjoint candidate blocks. On the full source chart, each block is good with conditional probability at least $p_* > 0$, up to a summable cluster correction. Hence*

$$\mathbb{P}(\text{fewer than } c_*\mu_\varepsilon \text{ good blocks}) \leq e^{-c\mu_\varepsilon}.$$

Conditioned on the exterior variables, the sum contributed by every good block has a $W^{s,1}$ density with uniform norm for any fixed s after enlarging m_0 finitely.

Proof. Separated insertion cells interact only through a convergent hard-core polymer correction. The finite-energy insertion estimate gives the uniform conditional lower probability. Exponential Chebyshev bounds for the block indicators, with the cluster correction absorbed into the exponent, give the first statement. On a good product patch, coarea gives a density with a uniform first Sobolev norm. Convolution of finitely many independent conditional block contributions increases the Sobolev order; bounded conditional density ratios from the polymer expansion preserve the uniform bounds. Thus every typical configuration, not merely one local component, contains linearly many smoothing blocks. $\square$

Theorem 1.4 (Full mixed lattice–continuous characteristic estimate). *Under the exact tilted law at $\lambda_{\varepsilon,H,a}$, the centered joint characteristic function of particle number and continuous constraints satisfies, uniformly in the source chart,*

$$|\varphi_\varepsilon(t, u)| \leq \begin{cases} e^{-c\mu_\varepsilon(t^2+|u|^2)}, & |t| + |u| \leq \delta, \\ e^{-c\mu_\varepsilon}, & \delta \leq |t| \leq \pi, \quad |u| \leq R, \\ e^{-c\mu_\varepsilon} + C_s(1 + |u|)^{-s}, & |u| > R, \end{cases}$$

for arbitrarily large fixed s .

Proof. The central estimate is the Taylor expansion at the exact finite saddle and Lemma 1.1. Away from the number lattice origin, a positive fraction of insertion cells yields the usual span-one factor $|1 + ze^{it}|/(1 + z) < 1$. For large continuous frequency, split according to the event in Lemma 1.3. Its complement is exponentially small. On the good event, integrate by parts across the linearly many Sobolev block densities; using only a fixed number of blocks already gives the prescribed polynomial order, while the remaining blocks preserve the bound. Thus no positive-mass singular remainder survives in the high-frequency estimate. $\square$

1.3. The admissible weighted shell. Let particle number be fixed at its lattice target. Let the continuous shell be

$$|Y - a'| \leq \delta_\varepsilon, \quad \delta_\varepsilon \downarrow 0, \quad \sqrt{\mu_\varepsilon} \delta_\varepsilon \longrightarrow \infty.$$

The target is the mean at the exact finite saddle.

Theorem 1.5 (Uniform exact-saddle shell coefficient). *Uniformly on compact regular source/target charts,*

$$\mathbb{P}_{H,\lambda_{\varepsilon,H,a},\varepsilon}\{N = N_{\varepsilon}, |Y - a'| \leq \delta_{\varepsilon}\} = \mu_{\varepsilon}^{-1/2}(c_{H,a} + o(1)),$$

where $0 < c \leq c_{H,a} \leq C$. In particular the logarithm of the shell probability is $o(\mu_{\varepsilon})$.

Proof. Use Fourier inversion in the exact lattice coordinate and smooth upper/lower approximations of the continuous box. The central region gives the joint Gaussian coefficient at the exact saddle. Since the continuous window contains a diverging number of central-limit standard deviations, its conditional Gaussian mass tends to one, while the exact number coefficient is of order $\mu_{\varepsilon}^{-1/2}$. The two minor arcs and the large continuous-frequency region are negligible by Theorem 1.4. Uniform covariance bounds give the positive coefficient bounds. $\square$

Theorem 1.6 (Source-dependent microcanonical pressure and prepared LDP). *Conditioning on the admissible shell yields*

$$Q^{\text{mc},a}(H) = Q(H, \lambda_{H,a}) - \lambda_{H,a} \cdot a - Q(0, \lambda_{0,a}) + \lambda_{0,a} \cdot a.$$

The initial empirical measure has the constrained grand-canonical entropy rate, and its Gaussian covariance is the Schur complement of the full constraint covariance. The theorem applies jointly to the B2 particle and actual-contact source class.

Proof. Change measure in the numerator to the exact finite saddle. On the shell the multiplier factor equals $e^{-\mu_{\varepsilon}\lambda_{\varepsilon,H,a} \cdot a + o(\mu_{\varepsilon})}$, and Theorem 1.5 makes the residual shell cost subexponential. Repeat with $H = 0$ for the denominator and pass to the normal pressure limit. Convex duality gives the constrained initial rate. Differentiating the saddle equation gives the covariance Schur complement. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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